

Costly Information and Beliefs: A More Direct Test of Rational Inattention

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Modern Models of Info Acquisition

In Theory:

$$\max_{\text{Info} \in I} E[\text{Revenue}|\text{Info}] - \text{Cost}(\text{Info})$$

- ✓ Decision variable: Info = Random Posteriors
(Distribution of Posteriors over State Space $\in \Delta(\Delta(\Omega))$)

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In the Lab:

Guess correct state $\implies$ win prize

- ✓ Observable variable: Choice given State (SDSC)
Approach: (Caplin and Dean (2015)) Reconstruct optimal Random Posteriors from SDSC for *Bayesian* agents

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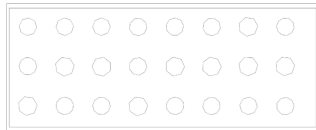
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Question: Is it WLOG that we use choice to test models of inattention? What is the appropriate choice variable?

What to ask in the Lab?



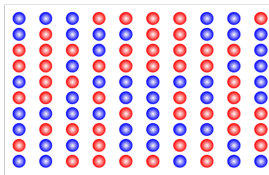
Dewan & Neligh (2020, JET)



Caplin et al (2020, QJE)

•With 50% probability there will be 4 correct equations.		
•With 50% probability there will be 3 correct equations.		
$13+25=38$	$4+19=23$	$37+21=56$
$29+32=58$	$24+25=60$	
$17+24=41$	$6+19=34$	

Dean & Neligh (2023, JPE)



Dean & Neligh (2023, JPE)

Figure: Perception Tasks

Possible choice variables:

- Direct Frame: “What is your intended **random posterior**?” vs.
- Natural Frame: “What is your optimal **information structure**?”

Main research questions:

1. How is information costly?

- Test models of RI **directly**, using Random Posteriors
("Direct Frame" Treatment)
- Test models of RI **indirectly**, using Information Structures
("Natural Frame" Treatment)

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1. How is information costly?

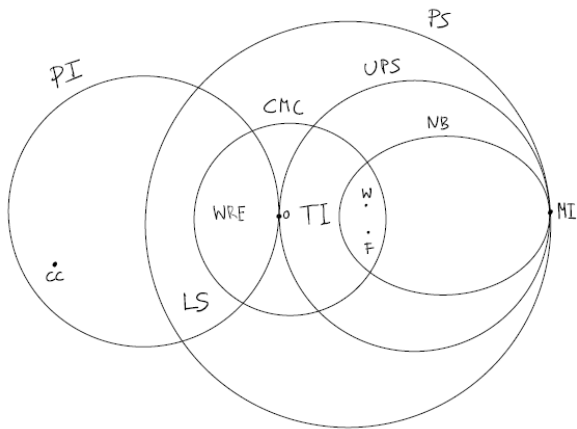
- Test models of RI **directly**, using Random Posteriors
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- Test models of RI **indirectly**, using Information Structures
("Natural Frame" Treatment)

2. Are the two frames of decision problems "practically" equivalent?

- Optimal random posterior = Optimal information structure
- Test if behavior from direct frame **extends to natural frame**
("Direct Frame vs. Natural Frame" Treatments)

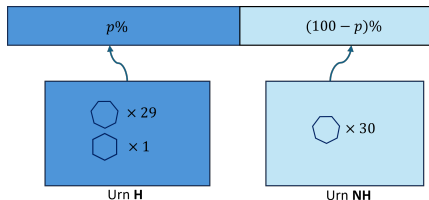
Models of RI

Strzalecki (2024)'s diagram



Design

Decision Problem



Decision Problem

- Task = Guess the Urn
- Two Urns: $\Omega = \{H, NH\}$
 - $p \equiv Pr(H)$: a prior belief
- Two Action: $A = \{nh, h\}$
 - $u(NH, nh) = u(H, h) = r \times u(\$20) > 0$
 - $u(NH, h) = u(H, nh) = 0$
- “Inspect the urn” before making guess
 - Draw polygons sequentially from the selected urn *without replacement*
 - Each polygon is shown for 0.7 sec

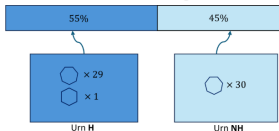
Design

Direct Frame Treatment

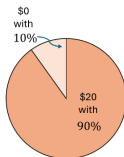
► More interfaces

ound 1 of 168

The chance of URN H being selected is 55%.
The chance of URN NH being selected is 45%.



Correct guess gives you 90% chance of winning \$20 .



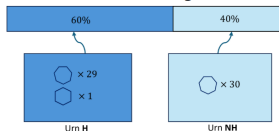
How likely do you want the guess NH to be correct in case no hexagon appears?

Design

Natural Frame Treatment

ound 1 of 168

The chance of URN **H** being selected is **60%**.
The chance of URN **NH** being selected is **40%**.



Correct guess gives you **50%** chance of winning \$20 .



How many polygons would you like to take out of the URN?

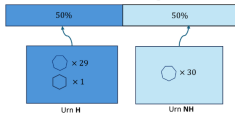
Next

Design

Natural+ Frame Treatment

Round 1 of 168

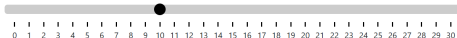
The chance of URN H being selected is 50%.
The chance of URN NH being selected is 50%.



Correct guess gives you 50% chance of winning \$20 .



Please click the bar below to reveal the slider to indicate the polygons to be taken out.



Taking out **10** polygons

The likelihood of the Guess NH to be correct in case no hexagon appears: **60.00%**

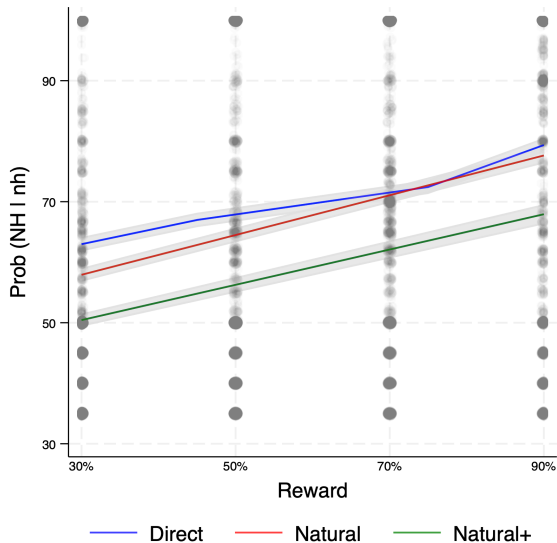
How many polygons would you like to take out of the URN?

Our Findings

Findings

- ① Most subjects are rationally inattentive.
- ② Cost is born differently depending on frame
 - Both UPS and PI are empirically valid, as well as the zero cost model.
 - Frame seemingly affects the dominant behavior.
- ③ Aggregate: More UPS-type in direct than natural frame, PI is the majority behavior across frames
- ④ Individual: significant portion switch behavior depending on frame, UPS-consistent type have high accuracy
 - A significant portion switches mostly from UPS to PI. The majority of subjects are consistent.
 - Frame affects accuracy regardless of consistency.

Rationally Inattentive? (Aggregate)



Rationally Inattentive? (Individuals)

$$\text{Prob}(\text{NH}|\text{nh}) = \beta_0 + \beta_1 \text{reward} + \beta_2 \text{prior} + \epsilon$$

	$\beta_1 \geq 0$	$\beta_1 < 0$	Total
Direct	50	0	50
Natural	40	1	41
Natural+	34	0	34

Table: Weakly Responsive

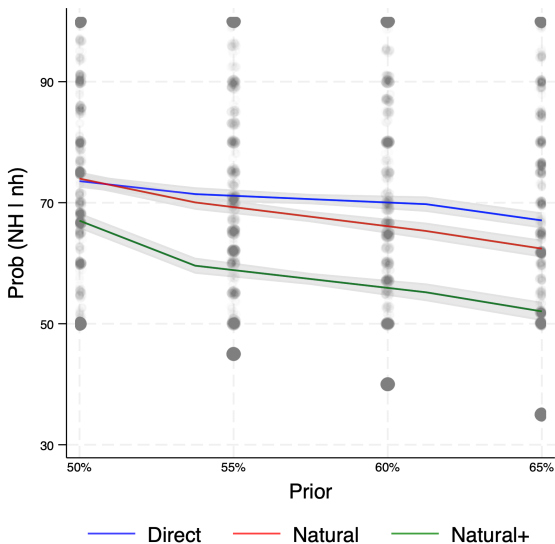
	$\beta_1 > 0$	$\beta_1 \leq 0$	Total
Direct	24	26	50
Natural	23	18	41
Natural+	19	15	34

Table: Responsive

Result 1:

Most subjects are weakly responsive to the reward. Close to half are strictly responsive.

How is Info Costly? (Aggregate)



How is Info Costly? (Individuals)

$$\text{Prob}(\text{NH}|\text{nh}) = \beta_0 + \beta_1 \text{reward} + \beta_2 \text{prior} + \epsilon$$

	$\beta_2 = 0$	$\beta_2 < 0$	Neither
Direct	25	15	1
Natural	11	24	1
Natural+	6	18	0
Total	42	58	1

Table: Prior Responsiveness

How is Info Costly? (Individuals)

$$\text{Prob}(\text{NH}|\text{nh}) = \beta_0 + \beta_1 \text{reward} + \beta_2 \text{prior} + \epsilon$$

	$\beta_2 = 0$	$\beta_2 < 0$	Zero	High	Neither
Direct	29	20	9	3	1
Natural	14	26	6	4	1 (Not RI)
Natural+	9	25	0	5	0
Total	52	71	15	12	2

Table: Model

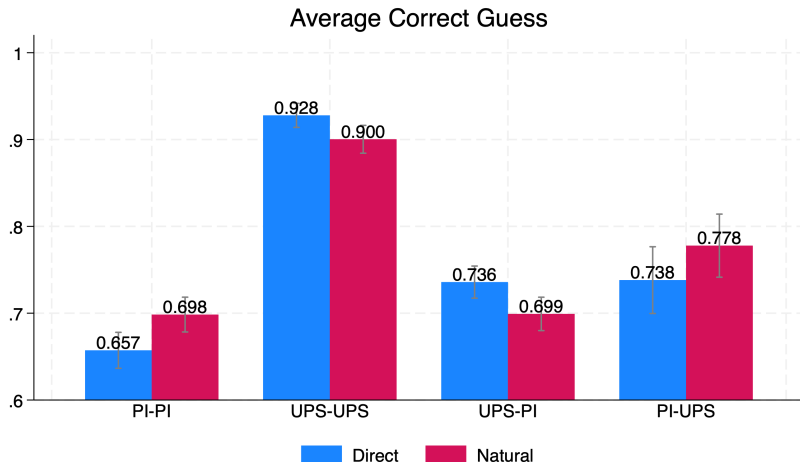
Result 2:

Both UPS and PI are empirically valid, as well as the zero cost model. Frame seemingly affects the dominant behavior.

Is Behavior Consistent? (Individual)

	Natural UPS	Natural PI
Direct UPS	9	14
Direct PI	5	12 (Non RI: 1)
Consistent	21	
Switch	19	

Is Behavior Consistent? (Aggregate)



Result 3:

The majority of subjects are consistent, though a significant portion switch mostly from UPS to PI. Frame affects accuracy regardless of consistency.

Summary

Goals

- ① Study how individuals bear information costs in decision-making
- ② Examining the validity and consistency of Uniform Posterior Separable (UPS) and Prior Invariant (PI) models of rational inattention

Findings

- ① Cost is born differently depending on frame
- ② Aggregate: More UPS-type in direct than natural frame, PI is the majority behavior across frames
- ③ Individual: significant portion switch behavior depending on frame, UPS-consistent type have high accuracy

Thank you for your attention!

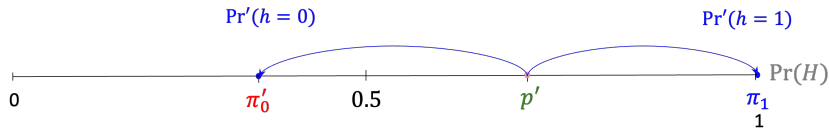
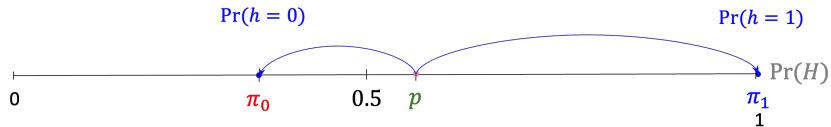
Bibliography I

- Caplin, A. and Dean, M. (2015). Revealed preference, rational inattention, and costly information acquisition. *American Economic Review*, 105(7):2183–2203.
- Strzalecki, T. (2024). *Stochastic choice theory*. Cambridge University Press.

Procedures

- Ohio State University students recruited via ORSEE
- Direct Frame: 50 subjects; Natural: 41; Natural+: 34;
- Payments: Randomly picked two rounds for their guess (20 USD)
- Average of 28.42 USD (including 10 USD show-up)
- Range: 30-120 mins
- Total observations: 16,968

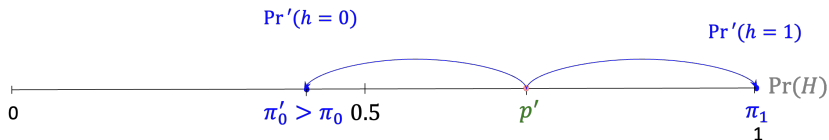
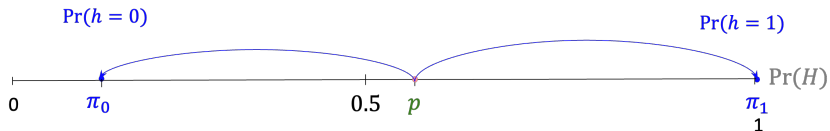
Varying Prior: UPS



UPS Test: $\pi_0|p = \pi'_0|p'$

[◀ Go back to "Choice of Informativeness"](#)

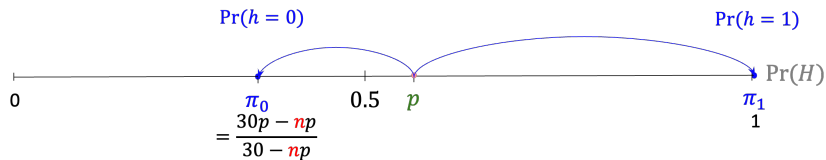
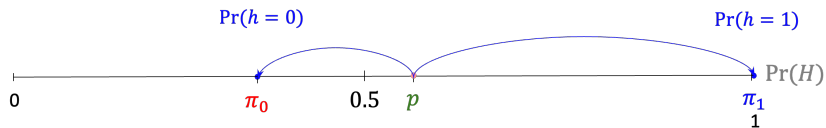
Varying Prior: PI



PI Test: $\pi'_0 | p' > \pi_0 | p$

[◀ Go back to "Choice of Informativeness"](#)

Varying Frames: Behavior Consistency



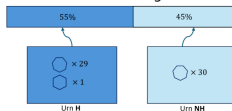
Consistency Test: $\pi_0|p = \pi_0|p$

◀ Go back to "Choice of Informativeness"

Interface 1/3

Round 1 of 168

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Next

Figure: Choosing Info

Interface 2/3

Round 1 of 168

Observe the polygons

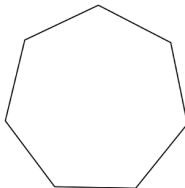


Figure: Drawing Info

Interface 2/3

Round 2 of 168

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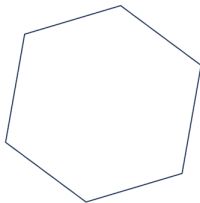


Figure: Drawing Info

Guess the URN

You guess that the selected Urn for this round is

☐ No Hexagon ☐ Hexagon

Next

Figure: Taking a Guess

◀ Go back to "Treatment Interface"